Machine learning applied to finance: an overview

Supervised, deep learning, generative AI, reinforcement learning: four families, four different uses.

PITFALL #1: OVERFITTING

Low error on training data + high error on test data = a model that memorized instead of learning

SUPERVISED ML

- Regression (predict a value) or classification (predict a category).
- Requires labeled data (we already know the historical right answer).
- The backbone of most scoring and forecasting models in finance.

DEEP LEARNING & NLP

- Deep neural networks for complex data (text, images, long series).
- NLP (language processing): sentiment analysis on news, earnings call transcripts.
- Needs a lot of data and computing power.

GENERATIVE AI & LLMS

- Generates text, summarizes documents, assists research and reporting.
- Hallucination risk: can produce false information with apparent confidence.
- Growing use in equity research, compliance and client service.

REINFORCEMENT LEARNING

- An agent learns by trial and error to maximize a reward (e.g. PnL).
- Used in order execution and algorithmic market making.
- Hard to audit and explain, so rare in production without strong guardrails.

Who uses what?

QUANT RESEARCHER
Supervised ML to build predictive signals and factors.

SYSTEMATIC TRADER
Reinforcement learning and deep learning for execution and pattern detection.

RISK / COMPLIANCE
NLP to monitor communications and detect risk signals in text.

PORTFOLIO MANAGER / ANALYST
Generative AI to speed up research, document summarization and reporting.

KEY TAKEAWAYS

1. A model that performs perfectly on training data is suspicious, not reassuring — always validate with cross-validation and out-of-sample testing.
2. The more complex a model (deep learning, RL), the harder it is to explain — a real challenge in regulated finance.
3. Generative AI is an assistance tool, not a source of truth: any output must be verified before being used for decisions.

Comparison table

FAMILY	GOAL	FINANCE USE CASE
Supervised	Predict a known value/category	Credit scoring, default forecasting
Unsupervised	Find hidden structure	Market regime clustering
Reinforcement Learning	Maximize a sequential reward	Optimal order execution
Generative AI	Generate/summarize content	Report summaries, assisted research

Real-world examples

CREDIT SCORING

A classification model predicts a borrower's default probability from their financial history.

REGIME CLUSTERING

An unsupervised algorithm automatically groups market days into regimes (calm, stress, trend) without any predefined label.

SENTIMENT ANALYSIS

An NLP model scores the tone (positive/negative) of earnings releases to anticipate a market reaction.

RL MARKET-MAKING AGENT

An agent learns to quote bid/ask prices that maximize profit while keeping inventory under control.

Quick glossary

OVERFITTING
A model that memorizes training data noise rather than learning a generalizable signal.

FEATURE ENGINEERING
Building relevant explanatory variables from raw data.

CROSS-VALIDATION
Technique to evaluate a model across several data splits and limit overfitting.

BACKTESTING
Testing a model or strategy on historical data before production use.

EMBEDDINGS
Dense numerical representation of text or data, used by NLP/LLM models.

HALLUCINATION
An LLM answer that sounds credible but is factually incorrect.